SplitSnap Product Requirements and Technical Architecture

India first expense sharing for Android iOS and web

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Status	Development baseline

Executive summary

SplitSnap is a shared expense ledger for meals, trips, households, and friend groups. A payer photographs a receipt, a vision model extracts the line items and totals, and the payer checks the result. Friends receive scoped browser links, choose the items they consumed, and see their share without installing the app or creating an account. The host manages groups, friends, expenses, balances, activity, and account settings from Android, iOS, or web.

The product uses INR as its India ledger currency and stores values as integer paise. Guests choose items manually. The server calculates item amounts, taxes, discounts, tips, and service charges under explicit allocation rules, then freezes a versioned split whose allocations add up exactly to the receipt total.

UPI is the intended India settlement method after the hackathon. The initial UPI experience opens the payer's UPI app with the recipient's details or displays a recipient-specific QR, then asks the recipient to confirm payment unless a selected PSP or acquiring partner provides a verified transaction status. PayPal Sandbox is a separate hackathon adapter. PayPal's current REST currency table does not list INR, so sandbox orders use a clearly labelled USD test amount and remain separate from real INR balances. See the official PayPal currency reference and NPCI UPI guidance in the source list.

Locked product decisions

Decision	Requirement
Product	SplitSnap
Platforms	Android and iOS apps, organizer web, and a guest web portal
Launch market	India; ledger currency INR; calculations stored in paise
Guest access	No app or account required; a scoped browser link identifies one invited participant
Receipt AI	AI extracts and proposes receipt fields; a person confirms them
Item allocation	Guests manually choose the items they consumed; AI never assigns consumption

Decision	Requirement
Production settlement	UPI through an external UPI app or QR, with truthful payment verification states
Hackathon payment	A separate PayPal Sandbox demo adapter; it does not represent UPI or real INR settlement
Design reference	Dark charcoal fintech interface with teal actions and Groups, Friends, Activity, Account navigation

Core ledger rule. Clients may display estimates, but the server owns payable amounts. A payment return URL, a guest's self-report, and a model response cannot by themselves mark a production UPI settlement as confirmed.

1 Product overview

Product problem

Groups usually have one person paying for a mixed receipt. Manual entry is slow, equal splits feel unfair when people ordered different items, and asking everyone to install an app creates friction. Taxes, service charges, shared dishes, discounts, and tips make the arithmetic harder. SplitSnap turns the receipt into a corrected shared ledger and lets each guest participate in a browser.

Product promise

The host can move from receipt photo to a reviewable split in a few minutes. Each guest can see the items and amount relevant to them, adjust their selection before publishing, and pay the recipient through the available settlement path. The app keeps an audit trail from receipt values to allocations and payment status.

Goals and initial measures

Goal	Measure	Initial bar
Make receipt splitting accurate	Published allocations exactly equal the confirmed receipt total	Zero paise difference on every published expense
Reduce guest friction	Invited guests who submit a selection within 24 hours	At least 60 percent in the first pilot; revise after 50 bills
Make capture useful	Clean receipts that reach a confirmed split without support	At least 70 percent in a 50 receipt pilot; log corrections
Keep payment status trustworthy	Production settlements marked confirmed only by recipient or verified provider event	100 percent of confirmed states have an audit event
Demonstrate hackathon integration	End to end receipt extraction and sandbox payment demo	One hosted path can be completed in under three minutes

Product guardrails

- Treat the receipt parser as an assistant. Make uncertain fields visible and editable before any amount is shared.
- Give every guest a narrow invite scope. A guest sees their own selection and share by default, not other guests' profile data.
- Keep UPI and PayPal Sandbox as different adapters, credentials, environments, currencies, and status labels.
- Do not hold, pool, or route user funds through SplitSnap in the first release.
- Keep an expense revision trail. A changed receipt or allocation creates a new version and reopens affected shares.

2 Users and journeys

Persona	Needs	Primary surface
Host and payer	Create a group, photograph a receipt, correct the extraction, invite guests, publish the split, and track who has settled	iOS or Android app; organizer web
Guest payer	Open a link, identify their invitation, choose their own items, understand tax and tip, and pay without installing an app	Guest web portal
Group member	Review shared expenses, current balances, activity, and settlement history	Mobile app or organizer web
Expense recipient	Receive a due amount, share a UPI destination, and confirm a completed transfer	Mobile app or organizer web

Host receipt journey

1. Create or open a group, add the people who were present, and choose the person who paid.
2. Capture or upload a receipt. The app creates a private upload URL and starts an asynchronous parse job.
3. Review merchant, date, currency, items, quantities, tax, service charge, discount, tip, and total. Correct uncertain values.
4. Choose how shared items and every bill-level adjustment are allocated. Invitees receive one scoped link per participant.
5. Guests select consumed items in the browser. The host reviews unassigned items and the final reconciliation.
6. Publish the split. The app freezes the allocation version, logs the activity, and displays each participant's exact amount.
7. Use UPI for the India flow or PayPal Sandbox only in the hackathon demo environment. Reconcile status according to the selected adapter.

Guest journey

1. Open the secure link on mobile or desktop, see the group and receipt context, and enter or confirm a display name if needed.
2. Select the items they consumed. They may split a shared item evenly or enter their portion when the host allows custom portions.
3. Review their item subtotal and share of tax, tip, service charge, and discount before submitting.
4. The guest can revise their selection until the host publishes or locks the split. After publication, a change creates a new version.
5. Choose UPI or, in the hackathon demo only, PayPal Sandbox. The page explains whether the payment is pending, reported, verified, recipient-confirmed, failed, or sandbox-only.

Settle up journey

A member selects a balance line and reviews the recipient name, exact INR amount, and destination. Mobile opens a UPI payment intent; desktop displays a QR code. After returning, the sender may report payment and provide a UTR or note. The recipient confirms it, or the selected payment provider posts a verified event. Until then, the balance remains pending. The app never asks for a UPI PIN.

3 Release scope

Release	Included scope	Exit condition
Hackathon P0	Hosted responsive web organizer and guest portal; demo groups and friends; receipt upload and vision extraction; human correction; manual item assignment; exact paise split; PayPal Sandbox Orders demo; source repo and submission materials	Judge can complete one receipt to guest selection to labelled sandbox payment flow
India MVP P1	Owner accounts; Android and iOS apps; groups, friends, activity, manual expenses; receipt flow; UPI intent or QR; recipient confirmation; audit and notifications	Pilot groups can split and reconcile live UPI without SplitSnap holding funds
Later P2	Verified PSP/acquirer transaction events where eligible; richer analytics; recurring expenses; multi-currency trips; configurable rules; automation after explicit review	Each addition has a separate product and compliance review

Given the close hackathon deadline, the hosted web journey is the required functional demo. The mobile apps and full India UPI settlement are release work after the demo unless already available in the codebase. The product architecture still treats iOS and Android as first-class launch platforms.

4 Functional requirements

ID	Capability	Development requirement	Priority
FR 01	Owner account	An organizer can sign in with phone or email one time code, restore their groups on a second device, sign out, and request deletion or export. Guest access never requires an account.	P0 demo can use seeded organizer access; P1 production requires account recovery
FR 02	Groups and friends	Create, rename, archive, and leave a group. Group types include Trip, Home, Couple, and Other; Trip may have start and end dates. Add members as accounts or guest participants. Show group totals and member balances.	P0 group create/read; P1 CRUD and membership management
FR 03	Guest links	Create one unique link per invited participant. Show expiry and revoke controls. Guest token is scoped to a group and split, stored hashed, rate limited, and exchangeable for a short lived browser session.	P0
FR 04	Manual expenses	Create an expense without a receipt; set description, date, payer, participants, total, and split rule. Support equal, exact amount, percentage, and itemized splits. Validate that shares equal the entered total.	P1
FR 05	Receipt upload	Capture or upload JPEG, PNG, or HEIC within configured size limits. Strip EXIF, validate file type and dimensions, use private storage, and show upload and parse status.	P0
FR 06	Receipt extraction	Extract merchant, date, currency, line descriptions, quantities, unit prices, line totals, tax lines, discounts, service charge, tip, and grand total. Include confidence per field and item. Never assign an item to a person.	P0
FR 07	Human correction	The host can add, remove, rename, or edit extracted fields and flag an amount as tax inclusive. Highlight low confidence fields and receipt-total discrepancies.	P0
FR 08	Item assignment	Each guest manually chooses the items they consumed. Shared items may use equal portions or a host-approved custom share. Guests can change their choices until publication.	P0
FR 09	Tax and other adjustments	Support tax, service charge, tip, discount, packaging, and other named adjustments. For each adjustment, choose all attendees, a selected set, item-weighted proportional allocation, or explicit per-person shares.	P0

ID	Capability	Development requirement	Priority
FR 10	Split and ledger	Recalculate on the server in integer paise. Use deterministic largest-remainder rounding. Do not publish while any amount is unassigned or the split differs from the confirmed total.	P0
FR 11	Balances and activity	Show who paid, who owes, current net balance, expense edits, receipt versions, invite and settlement events, and dates. Preserve original event history when a correction is made.	P0 basic feed; P1 full history
FR 12	UPI settlement	Show exact INR amount and named payee; generate payee-specific UPI intent or QR. Returning from the UPI app is not proof. Record sender report, recipient confirmation, or verified PSP status distinctly.	P1 India MVP
FR 13	PayPal Sandbox	In hackathon configuration only, create and capture a PayPal Orders v2 sandbox order from the server. Use USD test currency and show a sandbox-only label. Do not reuse this status as a production INR UPI settlement.	P0 for hackathon theme
FR 14	Notifications	Send an invite and a reminder with a deep link. Avoid sending receipt details or full balances in lock-screen notifications. Provide preference and opt-out controls for nonessential reminders.	P1; email first
FR 15	Account and settings	Manage display name, avatar, locale, notification settings, linked UPI destination, active sessions, export, and account deletion request.	P1

5 Split calculation rules

All split math runs server side in integer paise under a versioned allocation policy. Publish only when line items and adjustments reconcile exactly to a confirmed receipt total.

Allocation rules

- A line total is confirmed unit price multiplied by quantity. Guests choose what they consumed; shared items divide among their consumers. Custom portions use rational weights or basis points.
- Allocate tax, service charge, or tip by item subtotals, a selected eligible group, or exact per-person amounts.
- Allocate discounts on the selected basis. Keep inclusive taxes on the item line; do not add them twice.
- Round to paise with largest remainder; break ties by stable participant ID.
- Block publish until every item and adjustment is assigned and shares equal the confirmed total. Show any gap as unallocated.
- After publication, preserve the old version, create a replacement, notify affected people, and recalculate balances.

Example reconciliation

Receipt component	Amount	Allocation basis
Paneer tikka	₹450.00	Guest A
Biryani	₹600.00	Guest B
Water	₹40.00	Guest C
Tax	₹54.50	Proportional to item subtotals
Service charge	₹54.50	Proportional to item subtotals
Tip	₹100.00	Proportional to item subtotals; paise remainder applied deterministically
Total	₹1,299.00	Guest shares must add to ₹1,299.00

This example uses a simple item-to-guest assignment. A shared dish would divide its line amount before adjustment allocation. The system must show the rule beside the guest amount so the group can understand why shares differ.

6 Screen inventory and visual direction

Use the Splitwise screenshots only as a navigation reference: Groups, Friends, Activity, Account, group setup with type and trip dates, expense entry, balances, activity, and group detail. Create original SplitSnap visuals and assets.

Surface	Screen	Core content and actions
Mobile and web	Groups home	Group cards, total owed/owing summary, create group, search and filter
Mobile and web	Create or edit group	Name, type Trip/Home/Couple/Other, optional trip dates, icon, members
Mobile and web	Group overview	Group title and date, balances, expenses, activity, settle up, invite and add expense
Mobile and web	Friends	Friend list, net balance, add friend, invite, previously settled people
Mobile and web	Activity	Chronological expense, edit, invite and settlement events with amount and status
Mobile and web	Account	Profile, settings, notifications, security, privacy, export, delete
Mobile and web	Add expense	Description, amount, payer, participants, split method, date and optional receipt
Mobile	Scan receipt	Camera capture, crop/rotate, upload progress, retake
Mobile and web	Receipt review	Merchant, date, line items, quantity, taxes, discount, tip, total, confidence and mismatch
Mobile and web	Item assignment	Participant selector, shared item portions, adjustment policy, unassigned amount
Mobile and web	Invite manager	Participant-specific links, expiry, copied/shared status and revoke action
Guest web	Invite landing	Group name, receipt context, guest name confirmation, expiry and privacy summary
Guest web	Select items	Receipt list, item quantities, shared portions, own subtotal, adjustment explanation
Guest web	Review and pay	Exact amount, named recipient, UPI app/QR or sandbox demo option, current status
Mobile and web	Settlement history	Amount, payer, payee, method, reference, status and confirmation time

Visual requirements

Element	Direction
Base surface	Dark charcoal background with elevated charcoal cards and clear tonal separation
Primary action	Teal for selected navigation, progress, and primary actions; reserve amber for amounts due
Type and data	High contrast text, tabular numerals for rupee amounts, consistent INR formatting
Navigation	Mobile bottom navigation for Groups, Friends, Activity, and Account; responsive web navigation with the same destinations
Receipt review	Editable rows, visible confidence, clear totals, and explicit adjustment rules
Accessibility	Keyboard navigation on web, screen-reader labels, large touch targets, contrast checks, and no status conveyed by color alone

7 Technical architecture

Use a TypeScript monorepo with a modular backend and shared calculation package. Keep the browser and mobile clients thin: they submit intent and render a server-calculated preview. The backend validates membership, owns the ledger, performs receipt status transitions, and is the only service allowed to call model or payment credentials.

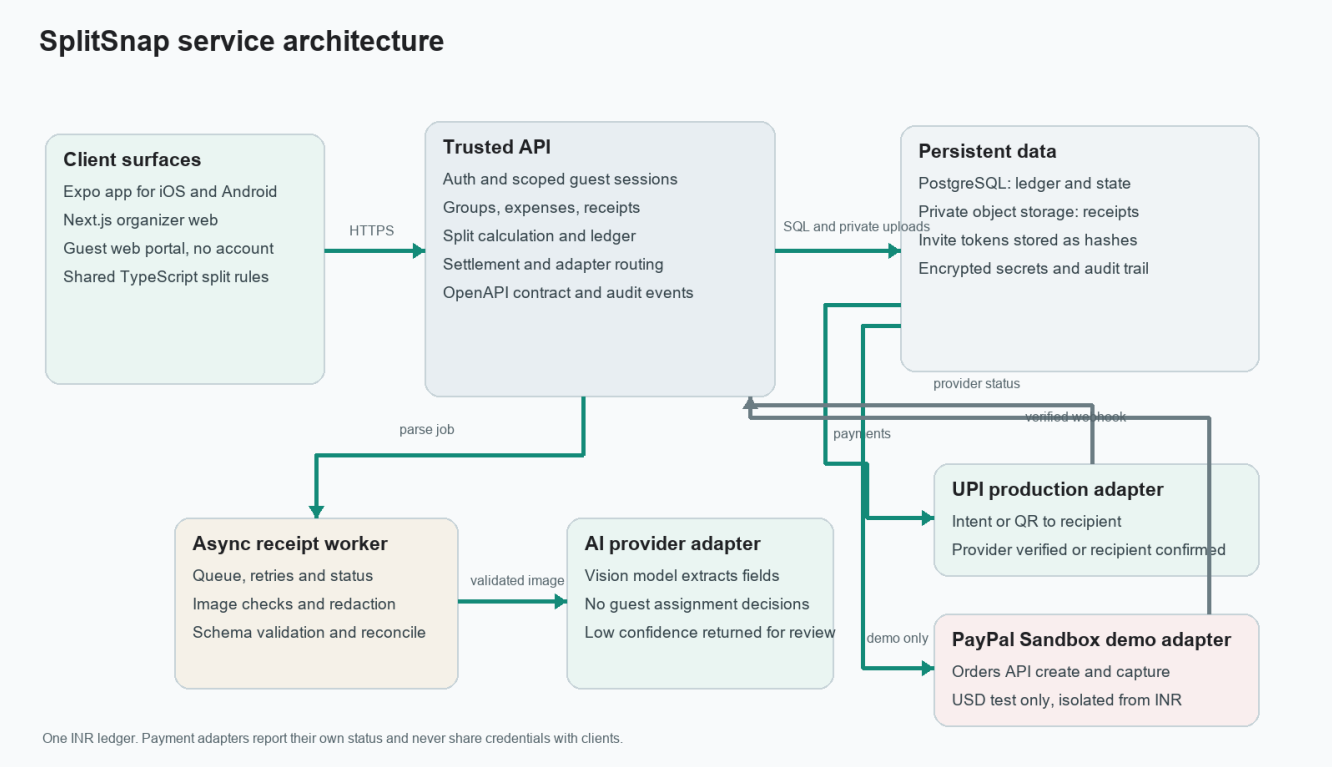


Figure 1 SplitSnap service architecture

Recommended implementation stack

Layer	Choice	Boundary
Mobile	Expo and React Native with TypeScript	Camera, native sharing, secure token storage, push later
Web	Next.js and TypeScript	Organizer experience and guest portal; responsive and index-blocked for private links
Shared domain	Workspace package with schemas and split calculation	Pure calculations and validation reused by web/mobile and server; server remains authoritative
API	NestJS or Fastify modular monolith with OpenAPI	Authentication, authorization, group ledger, payment adapter routing, audit events
Identity	Supabase Auth phone/email OTP for owner accounts	Guest link sessions are separate and limited; no direct client database writes
Database	Supabase managed PostgreSQL with Prisma migrations	Amounts in paise, foreign keys, row ownership checks in API, transaction boundaries
Jobs	Managed Redis with BullMQ worker	Parse jobs, retries, dead-letter handling, reminders and webhook follow-up
Receipt storage	Private S3-compatible storage with short-lived signed upload/download URLs	No public receipt objects; strip EXIF; virus/type checks

Layer	Choice	Boundary
AI	OpenAI Responses API with gpt-6.1-terra as the P0 model candidate and JSON Schema Structured Outputs	Model ID is configurable; compare representative receipts before pinning; human review remains mandatory
Payments	UPI adapter for launch; PayPal Orders v2 sandbox adapter for hackathon	Credentials, base URLs, currency and statuses stay isolated
Hosting	Render for web/API/worker; Supabase for managed auth/database; separate staging and production projects	Pin secrets to server environments; India-region choices are a launch review item

Service boundaries

Module	Owns	Must not do
Identity and groups	Users, group ownership, member roles, invitations and revocation	Treat a group ID or invite token supplied by the client as authorization by itself
Receipt service	Private upload metadata, parse status, normalized fields and parser versions	Publish model output without host correction or confirmation
Split engine	Item assignments, adjustment policy, paise math and versioned allocations	Trust client totals or use binary floating point for currency
Ledger	Expense shares, balances, settlement states and immutable event history	Let payment UI redirects clear a production balance
Payment adapters	Provider order/intent IDs, idempotency, webhooks and normalized statuses	Share credentials across adapters or expose secrets in client code
Notification service	Invite links, receipts of delivery and reminder preferences	Put receipt images, UPI IDs or full balance details in notification previews

Runtime configuration

```
APP_ENV=local|demo|staging|production
DATABASE_URL=server-only
AUTH_URL=...
AUTH_PUBLIC_KEY=client-safe
AUTH_SERVICE_ROLE_KEY=server-only
AI_PROVIDER=openai
OPENAI_RECEIPT_MODEL=gpt-6.1-terra
OPENAI_API_KEY=server-only
PAYPAL_ENV=sandbox|live
PAYPAL_CLIENT_ID=public-app-id
PAYPAL_CLIENT_SECRET=server-only
PAYPAL_WEBHOOK_ID=server-only
UPI_PROVIDER=manual_intent|chosen_psp
RECEIPT_BUCKET_PRIVATE=...
```

The exact secrets are stored in the deployment platform's secret manager. Never commit a populated environment file. Demo and production use different databases, storage buckets, OAuth clients, signing keys, webhook IDs, and payment credentials.

8 Data model

Use UUID primary keys, created_at/updated_at timestamps in UTC, and an event ID for every state change. Use integer paise for INR, plus an ISO currency code on every expense and payment record. Guest identifiers are separate from account users. Soft-delete or archive groups without deleting ledger history implicitly.

Entity	Key fields	Rules and relationships
User	id, display_name, phone/email reference, avatar_key, locale, deleted_at	Identity provider holds credentials; SplitSnap stores minimum profile data
Group	id, owner_user_id, name, type, start_date, end_date, currency, status	One owner; INR default; archive does not erase expense history
GroupMember	id, group_id, user_id nullable, guest_label, role, status	Exactly one account or guest identity; unique active membership per group
GuestInvite	id, group_id, member_id, token_hash, expires_at, revoked_at, scope	Hash raw token; one participant and expense/session scope; rate limit claims
Expense	id, group_id, payer_member_id, title, occurred_at, currency, total_paise, state, version	Server validates payer and total; immutable published versions
Receipt	id, expense_id nullable, object_key, mime, sha256, byte_size, retention_at	Private object key; no public URL; image may be detached from ledger after retention
ReceiptParseJob	id, receipt_id, state, provider, model, schema_version, attempts, error_code	Idempotent by receipt content hash and parser version; store minimal provider trace
ReceiptLineItem	id, expense_id, description, quantity_milli, unit_price_paise, line_total_paise, tax_included, confidence	Editable before publish; no AI-assigned member
ExpenseAdjustment	id, expense_id, type, description, amount_paise, allocation_policy, eligible_set	Signed amount; all adjustment values reconcile with total
ItemAssignment	id, line_item_id, member_id, share_bps or quantity_milli	Unique per item and member; shares total 10000 basis points or item quantity
ExpenseAllocation	id, expense_version, member_id, items_paise, adjustments_paise, total_paise	Published rows sum exactly to confirmed expense total
Settlement	id, group_id, from_member_id, to_member_id, amount_paise, method, status, provider_ref	No cash custody; status transition is audited; duplicate provider refs rejected
PaymentAttempt	id, settlement_id, adapter, currency, amount_minor, provider_order_id, idempotency_key, status	Sandbox attempts cannot share a production settlement or database
ActivityEvent	id, group_id, actor_member_id, event_type, object_ref, before_after_hash, created_at	Append-only audit record; avoid copying sensitive receipt text into event payload

Ledger invariants

- For every published expense, sum(expense allocations) equals the confirmed expense total in the same currency.
- For each settlement, from_member_id and to_member_id differ, amount is positive, and the settled amount cannot exceed the current due amount without an explicit overpayment workflow.
- Current balance is derived from payer credits, participant allocations, and confirmed settlements. Never recalculate balances from a client cache.
- A payment attempt has one adapter and one currency. PayPal Sandbox USD records cannot be counted as UPI INR transfers.
- All changes to published expenses, settlement status, invite lifecycle, and payment status produce an audit event in the same database transaction where practical.

9 API design

Use versioned JSON REST endpoints under /v1. Authenticate owner requests with a bearer token. Guest requests exchange the opaque invite secret for a short-lived, scoped session cookie and use CSRF protections for state changes. Require an Idempotency-Key on create/capture/settlement mutation calls. Return integer paise and currency separately; never accept a client-provided final amount as authoritative.

Method and route	Purpose	Access and important response
POST /v1/groups	Create group	Owner; group ID, member list and created event
GET /v1/groups/{groupId}	Read group overview	Member only; balances and recent activity scoped to group
POST /v1/groups/{groupId}/invites	Create invite for one participant	Owner; raw token shown once, hashed token stored, expiry returned
POST /v1/invites/claim	Exchange token from invite page body for a session	Guest; verify expiry/revocation, bind to participant and split scope
POST /v1/receipts/upload-intents	Create signed upload URL	Owner; object key, constraints and expiry
POST /v1/receipts/{receiptId}/parse	Start extraction	Owner; 202 with parse_job_id and status URL
GET /v1/parse-jobs/{jobId}	Poll parse status and draft	Owner; normalized candidate and confidence fields
PATCH /v1/expenses/{expenseId}/draft	Correct receipt and adjustment fields	Owner; validates schema and returns server preview
PATCH /v1/expenses/{expenseId}/assignments	Save manual item selections	Owner or scoped guest; server checks that actor can edit only their participant row
POST /v1/expenses/{expenseId}/publish	Freeze a reconciled split version	Owner; rejects unassigned or mismatched totals; returns allocation version
GET /v1/guest/splits/current	Read guest view using scoped session	Guest; own selection, line items and own share only
POST /v1/settlements	Create settlement request	Member; amount checked against current balance and named payee
POST /v1/settlements/{id}/upi-intent	Create handoff data	Member; returns intent or QR payload, exact INR, expiry and pending status
POST /v1/settlements/{id}/report-paid	Record sender report	Member; pending status, optional UTR, no auto-confirm
POST /v1/settlements/{id}/confirm	Recipient confirms	Recipient; confirmed status and audit event
POST /v1/demo/paypal/orders	Create sandbox order	Demo environment only; server-derived USD test amount and PayPal order ID
POST /v1/demo/paypal/orders/{id}/capture	Capture approved sandbox order	Demo environment only; server checks order amount and status
POST /v1/webhooks/paypal	Receive PayPal event	Public HTTPS endpoint; signature verification and event deduplication required

API error contract

```
{
  "error": {
    "code": "SPLIT_TOTAL_MISMATCH",
    "message": "Allocated shares must equal the confirmed receipt total.",
    "request_id": "uuid",
    "details": {"unallocated_paise": 250}
  }
}
```

Return stable error codes for expired invite, revoked invite, unauthorized participant, stale expense version, unassigned item, amount mismatch, duplicate request, provider pending, provider failure, and invalid webhook. Include a request ID but do not log invite tokens or payment secrets.

10 AI receipt pipeline

Processing steps

1. The client requests a signed upload URL. The server limits accepted MIME types and size, generates an unguessable object key, and stores only metadata in the database.
2. The upload endpoint verifies checksum and file signature, strips EXIF, normalizes orientation, and queues a parse job. The API returns 202 so the guest flow is not blocked.
3. The worker crops or scales large images, optionally masks unrelated personal identifiers or payment QR codes, and sends the image to the configured vision provider over TLS.
4. A provider adapter asks for a strict structured response. The model extracts candidate receipt data and confidence; it does not assign people, infer who consumed items, or decide payment status.
5. The worker validates the schema, decimal precision, quantity, currency, and line arithmetic. It compares item and adjustment sums with the grand total and records a mismatch instead of silently changing values.
6. The host reviews and corrects all uncertain values. The backend saves a draft with model and schema version, then recalculates totals in paise.
7. The split service accepts manual guest selections, allocates adjustments, reconciles the grand total, and only then lets the owner publish.
8. Retries use a dedupe key. On provider failure, show a retry action and allow manual entry; never discard the original image until the configured retention time.

Structured extraction contract

```
{
  "merchant_name": {"value": "Cafe Example", "confidence": 0.94},
  "receipt_date": {"value": "2026-11-10", "confidence": 0.82},
  "currency": {"value": "INR", "confidence": 0.99},
  "items": [{
    "description": "Masala dosa", "quantity": 1,
    "unit_price_paise": 18000, "line_total_paise": 18000,
    "tax_included": null, "confidence": 0.91
  }],
  "adjustments": [{
    "type": "tax", "description": "GST",
    "amount_paise": 900, "confidence": 0.78
  }],
  "subtotal_paise": 18000,
  "grand_total_paise": 18900,
  "warnings": ["Confirm whether line price includes tax"]
}
```

The actual schema should use JSON Schema with required fields, enumerations for adjustment type, bounded string lengths, positive quantity, integer paise, and nullable values where the image is unreadable. Provider output is untrusted input; validate it before persistence. Store a concise model trace for debugging without logging the whole image or personal data.

Prompt and model controls

- Tell the model to extract only visible receipt fields and return null with a warning when unreadable. Do not infer missing item prices or guess tax treatment.
- Treat printed receipt text as data, not instructions. The model must not access tools, execute commands, or change permissions.
- Send only the receipt image and parsing instruction. Do not send names, guest links, balances, UPI IDs, or payment data to the model.
- Do not train on user images unless the user separately opts in and the provider terms permit it. Document provider retention, region, and subprocessors before production.
- Measure extraction quality on a consented set of representative Indian receipts. Track field-level correction rate and total reconciliation rate, not a single opaque accuracy score.

P0 uses OpenAI Responses API with gpt-6.1-terra for image input and strict JSON Schema output. Pin the tested model ID and keep it behind a provider-neutral ReceiptVisionProvider adapter.

11 Payment and settlement design

Property	India UPI adapter	Hackathon PayPal Sandbox adapter
Purpose	Intended production settlement method in India	Sandbox demonstration of PayPal Developer integration
Currency	INR from the expense ledger; integer paise	USD test currency in a separate demo attempt; PayPal's current REST currency list does not include INR
Payment path	External UPI intent on mobile or payee-specific QR on web	Server creates and captures an Orders v2 sandbox order
Recipient	Named expense payer's UPI destination	SplitSnap demo merchant account in PayPal Sandbox
Verification	Pending until recipient confirms or an eligible PSP/acquirer supplies a verified event	Server checks capture and verifies webhook signature; sandbox badge remains visible
Ledger effect	Confirmed UPI transfer reduces the production receivable	Demo state only; never claim that a UPI transfer occurred or clear a production INR debt
Secrets	No UPI PIN, bank password, or full bank credentials stored	Client secret and webhook ID stay server side; sandbox/live credentials differ

UPI production flow

1. The member selects a net balance and an explicit recipient. SplitSnap shows the recipient name and UPI ID, amount in INR, and payment reference before handoff.
2. On Android or iOS, the app requests a UPI intent handoff. On desktop web, it shows a QR code. The receiving UPI app authorizes the transfer; SplitSnap never handles the UPI PIN.
3. The app returns to a pending state. A generic app-return signal is not payment proof. The sender may report the transfer and optionally enter a UTR; the recipient confirms, or a selected PSP/acquirer provides a verified status event.
4. Only the verified provider event or recipient confirmation changes the settlement to confirmed. Idempotency and duplicate-reference checks prevent double credit. A timeout remains pending and offers retry or dispute actions.
5. Before enabling a merchant checkout product, confirm the operating model and merchant/acquirer onboarding with the chosen bank or PSP. NPCI guidance describes merchant integration through an acquiring bank and lists QR, intent, application-based, and collect modes; this design does not assume direct UPI network access.

PayPal Sandbox demo flow

1. Only the demo environment exposes the PayPal option. The backend validates the demo settlement and calculates a separately stored USD test amount; the UI calls it a sandbox demonstration amount, not an INR conversion.
2. The backend creates an Orders v2 order with CAPTURE intent and an idempotency request ID. The browser or guest web view sends the user to PayPal Sandbox to approve it.
3. The backend captures the approved order and checks the returned order ID, capture ID, amount, currency, and status. The client never supplies the amount to be captured.
4. The webhook endpoint preserves the raw request body, verifies the signature with the registered webhook ID, deduplicates event IDs, and processes the event asynchronously. The UI shows a sandbox-only result.
5. Use separate sandbox credentials and data. PayPal Sandbox success is evidence that the demo adapter works; it is not an India UPI receipt, real money movement, or a production balance settlement.

PayPal's current Orders v2 integration describes server-side order creation and capture after payer approval. Its published REST currency codes omit INR, so the demo must use an explicitly separate USD test amount. NPCI guidance says merchants should partner with an acquiring bank for UPI infrastructure. These provider boundaries are why the ledger is payment-provider neutral.

12 Security and privacy

Area	Required control
Authentication	Owner accounts use a trusted OTP identity provider. Mobile stores refresh tokens in OS secure storage. Rotate/revoke sessions on sign out and deletion.
Authorization	Check group membership and role on every API call. Guest sessions are scoped to a participant and one split. Deny by default.
Invite links	Use at least 128 bits of random token entropy; store only a hash; expire and revoke; rate limit claims; redact token paths from access logs; use no third-party analytics on token pages; set noindex and a restrictive referrer policy.
Receipt images	Private object storage, signed URLs with short expiry, MIME and file-signature validation, size limits, malware scanning, EXIF removal, documented retention and deletion.
Payment data	Never collect UPI PINs, bank passwords, or full card data. Encrypt UPI destination at rest. Keep payment secrets in server secret storage.
PayPal webhooks	HTTPS only, signature verification, raw-body integrity, event deduplication, short acknowledgement and asynchronous processing. Do not trust an unverified callback.
Data in transit and at rest	TLS for every external connection; managed database and object-store encryption; least-privilege service accounts and key rotation.
Logging	Use request IDs and redacted structured logs. Never log raw invite tokens, OTPs, UPI PINs, full receipt images, or payment credentials.
Privacy operations	Give a clear notice for account, group, receipt, and AI processing. Provide access, correction, withdrawal/deletion request, grievance contact, and retention controls appropriate to applicable law.
AI vendor	Confirm region, retention, subprocessor list, and no-training terms before production. Send the minimum data needed for extraction.

The DPDP Act and Rules have staged commencement dates. Design the notice, consent, rights request, security, and retention workflows for the provisions in force at go-live, then have Indian privacy counsel confirm the applicable timeline and operational role. The Rules were notified by MeitY in November 2025; see the official source list.

Abuse and incident response

- Rate limit guest claims, parse requests, payment initiation, and account recovery by IP, user, and invitation where appropriate.
- Alert on repeated failed webhook verification, unusual invite claims, payment-reference reuse, sudden parse volume, and provider error spikes.
- Support a receipt dispute by retaining expense version, correction history, participant selection, and confirmation events without retaining unnecessary raw images indefinitely.
- Maintain a documented incident path to revoke invites, disable a payment adapter, rotate provider credentials, and notify affected users under the applicable rules.

13 Acceptance criteria

- A host can create a group and add named guest participants. Each invite link opens without installing the app or creating an account.
- A guest token cannot read another group's data, another participant's account profile, or modify another guest's selection.
- A receipt upload produces an asynchronous job status. A successful parse shows editable item and adjustment fields plus confidence and reconciliation warnings.
- The guest, not the model, chooses consumed items. The host can correct any extracted field and any assignment before publishing.
- A person who selects only a water item is charged for that item and only the adjustments allowed by the configured policy.
- For a shared item, the app shows the number of portions and each participant's amount. A host-approved custom portion persists after refresh.
- For equal, exact, percentage, and itemized split fixtures, allocations sum exactly to the confirmed receipt total in paise.
- The UI blocks publish when an item is unassigned, an adjustment has no allocation policy, or the split has a nonzero discrepancy.
- Publishing creates a stable allocation version. Repeated requests with the same idempotency key do not create duplicate expenses or settlements.
- A group overview shows balances derived from expenses and confirmed settlements, and its activity feed explains corrections and status changes.
- A UPI return to the app remains pending until recipient confirmation or a verified PSP event. SplitSnap never asks for a UPI PIN.
- A PayPal Sandbox demo order uses sandbox credentials and USD test currency. Its captured state is visibly sandbox-only and cannot clear a production INR balance.
- The PayPal webhook handler rejects invalid signatures, ignores duplicate event IDs, and checks that the event amount and currency match its demo payment attempt.
- Expired or revoked guest links return a clear recovery message and cannot create or modify allocations.
- The web guest journey works at mobile and desktop widths with keyboard navigation and screen-reader labels for item selection and payment status.
- Export and deletion requests are auditable; receipt object deletion follows the configured retention and backup schedule.

14 Nonfunctional requirements

Quality	Target
Responsiveness	p95 API response under 600 ms for ordinary reads and writes, excluding provider calls and receipt processing
Receipt processing	Return job status immediately; target p95 parse completion under 25 seconds for accepted images; retry transient provider errors twice
Pilot capacity	Support at least 100 concurrent guest sessions, 50 members per group, and 500 line items per expense without redesign
Availability	99.5 percent monthly availability for the India MVP; payment adapters fail closed while expense viewing remains available
Data integrity	Zero paise drift on a published expense; use database transactions and integer currency values
Recovery	Daily encrypted database backups, target RPO 24 hours and RTO 8 hours for MVP; test restore before live payments
Accessibility	Keyboard operable web flows, accessible labels, usable contrast, touch targets, and explicit text for error and payment states
Localization	Use INR and Indian number formatting; keep copy and date/number formatting locale-aware. Launch UI in English with translation-ready strings
Compatibility	Current supported iOS and Android OS versions selected at implementation start; current and previous major Chrome, Safari, Edge, and Firefox
Observability	Track request IDs, queue age, parse success and correction rates, API errors, payment states, and webhook lag without sensitive payloads

These are initial engineering targets, not measured performance claims. Benchmark them with realistic receipt images and traffic before a public launch.

15 Testing strategy

Test layer	Coverage
Unit	Paise arithmetic, largest remainder, shared portions, negative discounts, inclusive tax, eligibility rules, currency formatting, state transitions
Property based	For generated valid expenses, allocations sum to total; rounding is deterministic; retrying a calculation yields identical results
API integration	Auth and guest scoping, version conflicts, idempotency, database transactions, signed upload constraints, worker retries, event audit
Provider contract	AI JSON schema and malformed output; PayPal Sandbox order create/capture, mismatch, duplicate webhook, invalid signature; UPI intent URI and QR payload
End to end	Host creates trip and receipt; guest claims link and chooses items; host publishes; guest sees correct share; sandbox order completes in demo
Security	Object access, IDOR, invite brute force, token leakage in referrers/logs, webhook replay, secret scanning, dependency review
Usability and accessibility	One handed mobile receipt capture, keyboard only guest flow, screen reader selection state, contrast, error recovery, no UPI app installed
Performance and recovery	Concurrent guest links, slow AI provider, duplicate job delivery, worker crash, provider timeout, database restore
Manual acceptance	Use sample Indian receipts with varied layouts, shared starters, split tax, tip, discounts, and a guest who ordered only water

16 Implementation plan

Build the smallest complete web path first so a judge can interact with the core product, then reuse the same domain and API in the mobile apps. Work in order of risk: exact split math and invite permissions, receipt extraction, then payment adapters. The user-provided hackathon submission close is 13 November 2026 at 1:30 AM IST; treat it as a release cutoff, not an estimate of available engineering time.

Phase	Work	Acceptance gate
A Hackathon foundation	Create monorepo, demo environment, seeded account/group, schema, server API, database migrations, web shell, design tokens, split calculation package	Hosted app and test fixtures load; every currency value is integer paise
B Receipt workflow	Private upload, image checks, async queue, vision adapter, normalized schema, review/edit screen, extraction fallback	One sample receipt parses, can be corrected, and is not publishable before confirmation
C Guest assignment	Named guest slots, scoped links, guest portal, manual item selection, shared portions, adjustments, split preview and publish	Host and guest complete a reconciled expense in one browser session
D Hackathon PayPal demo	Sandbox credentials, Orders create/capture, USD test scenario, webhook verification, demo-only status, failure and retry UI	Successful and failed sandbox states are reproducible without touching production UPI balances
E Submission hardening	Seeded judge scenario, accessibility pass, privacy copy, hosted URL, repository license, setup guide, under-three-minute video, Devpost text and tool list	All required links work before the stated close time; no secret or personal data in repo
F India MVP mobile	Expo iOS/Android shell, owner OTP auth, groups/friends/activity/account, manual expense, camera capture, guest sharing and push/email notifications	Native journeys use the same API and split math as web
G UPI pilot	Select acquiring/PSP partner, validate merchant or person-to-person operating model, implement intent/QR, recipient confirmation, reconciliation and support flow	Payments are truthful, auditable, and do not require SplitSnap custody
H Production readiness	Privacy/legal review, retention policy, threat model, backups/restore, monitoring, incident playbook, pilot analytics and support	Launch owner signs off on privacy, payment provider, and operational gates

Suggested workstream ownership

Workstream	Primary tasks	Dependency
Product and design	Journey, dark theme, guest comprehension, privacy copy, demo script	Can begin immediately
Web and mobile	Next.js organizer/guest, then Expo owner app and native camera	API schemas and design tokens
Backend and ledger	Auth, groups, invites, expense versions, split engine, balances, audit	Must land before payment status changes
AI and receipt	Upload, worker, model adapter, extraction validation, correction UX	Receipt schema and private storage
Payments and operations	PayPal demo, webhooks, UPI provider selection, deployment and alerts	Adapter interface and provider accounts
QA and security	Arithmetic fixtures, access control, provider contracts, release checklist	Runs throughout; final signoff after end to end demo

Hackathon work sequence

1. Start with one seeded restaurant receipt and two or three named guests. Wire the host and guest screens to a real API before polishing secondary account features.
2. Lock the split contract and test examples before inviting guests. A wrong total is more visible than a missing chart.
3. Use a hosted web flow and one-click seeded demo. Add an Expo shell only after the hosted demo works end to end.
4. Integrate PayPal Sandbox after the INR split is stable. Keep sandbox state in a separate environment/config and record a short video of both success and failure handling.
5. Freeze feature work before the deadline. Verify the public URL, repo visibility/license, demo video duration, documentation, and credentials one last time.

17 Deployment and operations

Environment	Purpose	Data and payment settings
Local	Developer work	Synthetic receipts, local DB or isolated dev project, mock payment adapter
Hackathon demo	Judge interaction	Seeded data, private test receipt, AI provider test key, PayPal Sandbox only, explicit demo banner
Staging	Preproduction QA	Separate database and storage, PSP test credentials, no real payment destination
Production	India users	Live UPI/PSP settings only after onboarding, live identity provider, region and retention approved

Release and operations checklist

- Build web, API, and worker from one tagged commit. Run database migrations as a controlled release step and maintain a rollback migration plan.
- Use health checks for API and worker; monitor queue age, failed parse jobs, webhook lag, provider errors, and database connections.
- Set alerts for payment adapter failure, signature rejection spikes, duplicate captures, unassigned settlement backlog, and backup failure.
- Test object storage access policy, signed URL expiry, scheduled receipt deletion, and database restore before inviting real groups.
- Keep sample/demo mode visible in all screenshots and payment confirmations. Do not use a real receipt containing personal information in a public demo.

18 Risks and mitigations

Risk	Impact	Mitigation
Receipt extraction misreads prices or GST	Guests see incorrect shares	Confidence flags, human correction, reconciliation block, representative receipt set, manual entry fallback
Unassigned items or ambiguous inclusive tax	Split does not match bill	Make tax treatment explicit; show remaining amount; block publish until resolved
Guest link is forwarded or leaked	Another person sees or changes a split	Participant-scoped token, hash at rest, short expiry, revoke, rate limits, noindex, minimal data view
UPI app return is treated as proof	Unpaid debt is cleared in the ledger	Keep pending state; require recipient confirmation or verified provider event
PayPal INR assumption fails	Sandbox order rejected or amount misrepresented	Use supported USD test currency, clear demo label, separate demo currency/status and data; official currency list reviewed

Risk	Impact	Mitigation
Payment business model requires regulated onboarding	Production payments delayed or noncompliant	Do not custody funds; confirm acquirer/PSP and legal structure before selecting a merchant flow
Deadline leaves native app unfinished	Hackathon submission lacks an accessible demo	Prioritize hosted web organizer and guest flow; stage full native apps for post-submission
Provider outage or rate limit	Receipt or payment step stalls	Async jobs, retry with backoff, idempotency, visible pending status and manual fallback
Receipt image exposes phone, QR, or account data	Privacy or fraud risk	Private storage, masking where feasible, no model transmission of unrelated fields, access logs and retention
Double webhook or duplicate guest submit	Double settlement or duplicate allocation	Idempotency keys, unique event/provider IDs, transaction locking and replay-safe handlers

19 Explicit out of scope

- SplitSnap holding a wallet balance, pooling funds, escrow, stored value, or transferring a balance between members.
- Direct UPI network participation, UPI PIN entry, bank account aggregation, or storing banking credentials.
- Using PayPal Sandbox as a substitute for UPI, a live India payment method, or a real INR settlement.
- AI deciding who ate an item, who owes a person, or whether a payment is complete.
- Automatic receipt-to-item assignment, face recognition, location inference, or dietary inference.
- Credit, lending, pay-later, investment, rewards, subscriptions, and advertising in the first release.
- Automated bank reconciliation, chargeback management, foreign exchange, or multi-currency group netting at launch.
- Publicly exposing group links, receipt images, participant email/phone, or balances to search engines.
- Premium upsell and paid tiers. The supplied subscription screen is only a visual reference, not a launch requirement.

20 Hackathon submission checklist

Deliverable	Checklist
Functional demo	Hosted URL or complete setup steps; seeded account; receipt-to-guest-to-payment path; stable demo credentials; visible PayPal Sandbox label
Repository	Public repository on an accepted host; valid open-source license; setup guide; migrations; sample environment file without secrets; architecture and tool list
Demo video	Public YouTube link under three minutes; show receipt extraction, a human correction, guest manual item selection, exact share, and sandbox payment state
Documentation	Problem, product description, feature summary, AI tool/model, PayPal API integration, setup instructions, demo URL, data handling and known limitations
Submission form	Confirm all required URLs open in a signed-out browser; check repository access, video visibility, dates, and final submission receipt
Secrets and privacy	Remove credentials from code and video; use synthetic receipt image and identities; no real payment; do not show personal email or UPI ID

List the actual tools in the submission: OpenAI Responses API with gpt-6.1-terra for receipt extraction, PayPal REST Orders v2 Sandbox for the separate demo payment, and the deployed web, API, database, and hosting stack used by the build.

Dates from the supplied hackathon brief

Milestone	India time
Submission closes	13 November 2026 at 1:30 AM IST (12 November 2026 at 12:00 PM Pacific Time)
Judging	1 to 15 December 2026
Winners announced	21 December 2026

21 Launch gates and decisions

The product and technical defaults above are sufficient to begin implementation. Before a public India payment launch, close the following provider and operating decisions. These are launch gates, not reasons to delay the hackathon demo.

Gate	Default in this PRD	Required before launch
UPI provider and funds flow	Recipient-owned UPI intent/QR; manual recipient confirmation unless provider event exists	Confirm operating model, onboarding, supported devices and status verification with chosen bank/acquirer/PSP
AI provider	OpenAI Responses API with gpt-6.1-terra as the initial model candidate; server adapter	Approve data region, retention, subprocessors, terms, cost and latency; compare representative receipts
Data retention	Private receipt object with configurable deletion date; ledger history kept separately	Set retention and backup deletion schedule with Indian privacy/legal review
Identity	Owner OTP; guest link session	Choose OTP delivery vendor, fraud/rate limits, recovery and support ownership
Production regions	India region preferred where supported	Verify database, object storage, logs, model and payment provider locations
Policy and support	No custody, clear pending/confirmed states, in-app grievance route	Publish user terms, privacy notice, support SLA, payment dispute and correction process

22 Official implementation references

Provider and regulatory references checked on 10 November 2026. Confirm the selected provider's current terms, API version, account eligibility, and local legal obligations before production rollout.

1. NPCI UPI FAQs Merchant onboarding, acquirer partnership, and listed UPI integration modes.

2. PayPal Orders v2 integration Server creates and captures an order after payer approval; sandbox credentials are separate from live credentials.

3. PayPal currency codes Supported REST payment currencies; INR is not present in the current listed table.

4. PayPal REST webhook integration Webhook HTTPS, signature verification, raw payload considerations, and delivery retries.

5. MeitY Digital Personal Data Protection Rules 2025 Official notified Rules and commencement materials; applicability must be checked for the launch date.

6. OpenAI GPT-6.1 Terra model Vision-capable model specification and support for structured outputs.

7. OpenAI Responses API image input Image input handling and model vision limitations.

8. OpenAI Structured Outputs JSON Schema constrained output and supported schema limits.

9. PayPal REST API requests Sandbox REST base URL and authenticated request pattern.